Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Order of Operations in Algebraic Expressions

When an algebraic expression mixes different operations, you cannot simply work from left to right. You must follow a fixed order. This order is often remembered as **BODMAS** (or BIDMAS):

- **B**rackets first
- **O**rder (powers) — though you won't need them here
- **D**ivision
- **M**ultiplication
- **A**ddition
- **S**ubtraction

Division and multiplication are done before addition and subtraction. So in an expression like $18 - 6 \times 2$, you must multiply first: $6 \times 2 = 12$, then subtract: $18 - 12 = 6$.

The same rules hold when the terms are algebraic. For example, in $17y - 5 \times 2y$, multiply first: $5 \times 2y = 10y$, then subtract: $17y - 10y = 7y$. You combine the like terms only after doing the multiplication or division that comes first.

Example 1. Find the value of each expression.

(a) $18 - 6 \times 2$

Do the multiplication first: $6 \times 2 = 12$.

$18 - 12 = 6$

(b) $6 - 18 \div 3$

Do the division first: $18 \div 3 = 6$.

$6 - 6 = 0$

(c) $6 \times 2 - 2 - 40 \div 4$

Do the multiplication and division first, in any order: $6 \times 2 = 12$ and $40 \div 4 = 10$.

$12 - 2 - 10 = 10 - 10 = 0$

Example 2. Simplify the following expressions.

(a) $3x \times 2 + 5x$

Do the multiplication first: $3x \times 2 = 6x$.

Now add the like terms: $6x + 5x = 11x$

(b) $17y - 5 \times 2y$

Do the multiplication first: $5 \times 2y = 10y$.

Now subtract the like terms: $17y - 10y = 7y$

Example 3. Simplify the following expressions.

(a) $7x \div x + 5$

Do the division first: $7x \div x = \frac{7x}{x} = 7$.

Now add: $7 + 5 = 12$

(b) $4x - (6 \div 3)$

Work inside the brackets first: $6 \div 3 = 2$.

Now subtract: $4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Work inside each bracket first: $24x \div 6 = 4x$ and $1 \times 5x = 5x$.

Now combine the like terms: $4x + x + 5x = 5x + 5x = 10x$

Practice

1. Simplify: $4n + 3n \times 10$
2. Simplify: $32 + 8x \div 2$
3. Simplify: $3 \times 5x + 4x \div 2$
4. Simplify: $36y \div 6 + y + 2 \times 4y$
5. Simplify: $7 \times 3a - (3a + 5a) \times 2$

Section 2: Removing and Simplifying Brackets in Algebraic Expressions

When an algebraic expression has brackets, you can often make it simpler by removing the brackets and then collecting the like terms. Before you do anything else, always try to simplify the terms *inside* the bracket first.

What happens when you remove a bracket depends on the sign that stands just before it:

- A + sign before a bracket does **not** change the signs of the terms inside the bracket when you remove it.
- A − sign before a bracket **changes every sign** inside the bracket: every + becomes − and every − becomes + when you remove the bracket.

So the bracket in $a + (b + c)$ comes after a +, and removing it leaves the signs untouched. The bracket in $2a - (3b - c)$ comes after a − sign, so the + before $3b$ becomes − and the − before c becomes +.

Example 1 — removing a bracket after a + or a − sign

Simplify (a) $a + (b + c)$, (b) $a + (b - c)$, (c) $2a - (3b - c)$.

Solution

(a) The + before the bracket leaves the signs unchanged, so
$$a + (b + c) = a + b + c.$$

(b) Again the + before the bracket changes nothing:
$$a + (b - c) = a + b - c.$$

(c) The − before the bracket changes every sign inside: $3b$ becomes $-3b$ and $-c$ becomes $+c$, so
$$2a - (3b - c) = 2a - 3b + c.$$

Example 2 — removing brackets and collecting like terms

Simplify (a) $a + (a - b)$, (b) $6x + (3 - x)$, (c) $(3a + 2) + (5a + a)$.

Solution

(a) Remove the bracket, keeping the signs (it follows a +):
$$a + (a - b) = a + a - b = 2a - b.$$

(b) Remove the bracket, then group the like terms together before adding:
$$6x + (3 - x) = 6x + 3 - x = 6x - x + 3 = 5x + 3.$$

(c) Two brackets, both after a +. Remove them both, then collect the a terms:
$$(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 3a + 5a + a + 2 = 9a + 2.$$

Example 3 — removing a bracket after a − sign

Simplify (a) $5x - (3y + z)$, (b) $10d - (9c - 4d)$, (c) $(6x - y) - (4x - 5y)$.

Solution

(a) The − before the bracket changes $3y$ to $-3y$ and z to $-z$:
$$5x - (3y + z) = 5x - 3y - z.$$

(b) The − changes $9c$ to $-9c$ and $-4d$ to $+4d$:
$$10d - (9c - 4d) = 10d - 9c + 4d = 10d + 4d - 9c = 14d - 9c.$$

(c) Here the second bracket follows a − sign, so every sign inside it changes:
$$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 6x - 4x + 5y - y = 2x + 4y = 2(x + 2y).$$

Practice

1. Simplify $(12a - 7) + (4 - 5a)$.
2. Simplify $(8x - 2y) - (9x - 4y)$.
3. Simplify $12 - (7 - 2f) - (3 - f)$.
4. Simplify $(8a - 5) + (14 - 3a)$.

Section 3: Word Problems with Algebraic Expressions

A word problem puts a story around an algebraic expression. Your job is to read the story, turn each sentence into part of the expression, and then simplify. Words like "less than" mean subtract, and words like "more than" mean add. Start by deciding what each letter stands for, write the expression for the thing asked about, then simplify it by collecting like terms.

Example 1 — A pen costs ₦P and a pencil costs ₦15 less than a pen.

(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

Solution

The pen costs ₦P.

(a) A pencil costs 15 naira less than the pen, so it costs ₦(P − 15).

(b) A pen and a pencil together cost:
₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15).

Example 2 — A pencil costs ₦x and a rubber costs ₦40 more than a pencil.

(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

Solution

The pencil costs ₦x.

(a) The rubber costs 40 naira more than the pencil, so it costs ₦(x + 40).

(b) A pencil and a rubber together cost:
₦x + ₦(x + 40)
= ₦x + ₦x + ₦40
= ₦(2x + 40).

Example 3 — In a class there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave. How many students are now in the class?

Solution

Total number = number of boys + number of girls.

Number of boys now = x + 5 (5 join, so add).
Number of girls now = y − 3 (3 leave, so subtract).

Total number = x + 5 + y − 3
= x + y + 5 − 3
= x + y + 2.

So there are now x + y + 2 students in the class.

Practice

1. A pen costs ₦h and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.

2. Envelopes cost ₦d per pack, and a notepad costs ₦20 more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.
3. A pen and a pencil altogether cost ₦x. If a pen costs ₦y, find the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

Order of Operations in Algebraic Expressions

1. $34n$

   Do the multiplication first: $3n \times 10 = 30n$. Now add the like terms: $4n + 30n = 34n$.
2. $32 + 4x$

   Do the division first: $8x \div 2 = 4x$. Now add: $32 + 4x$. Since 32 is a number and $4x$ is a term in x, they are not like terms, so the answer stays $32 + 4x$.
3. $17x$

   Do the multiplication and division first: $3 \times 5x = 15x$ and $4x \div 2 = 2x$. Now add the like terms: $15x + 2x = 17x$.
4. $15y$

   Do the division and multiplication first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. Now add the like terms: $6y + y + 8y = 15y$.
5. $5a$

   Work inside the bracket first: $3a + 5a = 8a$. Now the expression is $7 \times 3a - 8a \times 2$. Do the multiplications: $7 \times 3a = 21a$ and $8a \times 2 = 16a$. Now subtract: $21a - 16a = 5a$.

Removing and Simplifying Brackets in Algebraic Expressions

1. $7a - 3$

   Remove both brackets (each follows a +, so the signs are unchanged): $12a - 7 + 4 - 5a$. Collect the like terms: $12a - 5a - 7 + 4 = 7a - 3$.
2. $-x + 2y$

   The second bracket follows a − sign, so change every sign inside it: $8x - 2y - 9x + 4y$. Collect like terms: $8x - 9x - 2y + 4y = -x + 2y$.
3. $2 + 3f$

   The two brackets each follow a − sign, so change every sign inside each: $12 - 7 + 2f - 3 + f$. Collect the number terms and the f terms: $12 - 7 - 3 + 2f + f = 2 + 3f$.
4. $5a + 9$

   Both brackets follow a +, so remove them with signs unchanged: $8a - 5 + 14 - 3a$. Collect like terms: $8a - 3a - 5 + 14 = 5a + 9$.

Word Problems with Algebraic Expressions

1. (a) ₦(h − 20); (b) ₦h + ₦(h − 20) = ₦(2h − 20).

   The crayon costs 20 naira less than the pen, so it costs ₦(h − 20). A crayon and a pen together: ₦(h − 20) + ₦h = ₦(h − 20 + h) = ₦(2h − 20).
2. ₦(2d + 20)

   A notepad costs ₦(d + 20). Total = ₦d + ₦(d + 20) = ₦(d + d + 20) = ₦(2d + 20).
3. ₦(3x − 3y), or ₦3(x − y)

   One pencil costs the total minus the pen, which is ₦(x − y). So 3 pencils cost 3 × ₦(x − y) = ₦3(x − y) = ₦(3x − 3y).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-54777ece	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____